MECHANICAL DETAILS:

The Process: After finalising the idea, we brainstormed on the functioning components we would require, their placements, and how they would look.

Starting with the target box, we ideated on multiple ways to incorporate an obstacle/circular wheel. Its size corresponds to the size of the box on which it rotates. We then looked for simple shooting mechanisms using different materials; first, a 3D print using filaments, and second, using MDF.

To have a moving turret in both the horizontal and vertical planes, we explored different forms until we reached an efficient structure that consumed fewer materials and functioned smoothly with minimal error.

The files for laser-cutting were done on LibreCAD, which excels in 2D design, enhancing accuracy in measurement and finish.

The 3D shooting mechanism was inspired by a YouTube video on "3D printed shooter ideas," which also helped us understand how we could build the same using MDF.